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In [1]: import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO

# Load pre-trained YOLOv8 model
model = YOLO("yolov8n.pt")

# Load image
image_path = "london-street-scene-cars-bikes-uk-august-bicycles-people-walking-shop"
image = cv2.imread(image_path)

if image is None:
    print("Image not found!")
else:
    print("Image loaded successfully!")

    # Object detection
    results = model(image_path)

    # Draw bounding boxes
    result_image = results[0].plot()

    # Display result
    plt.figure(figsize=(10, 7))
    plt.imshow(cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.title("YOLOv8 Object Detection")
    plt.show()

    # Print detected objects
    print("\nDetected Objects:")

    for box in results[0].boxes:
        class_id = int(box.cls[0])
        confidence = float(box.conf[0])
        class_name = model.names[class_id]

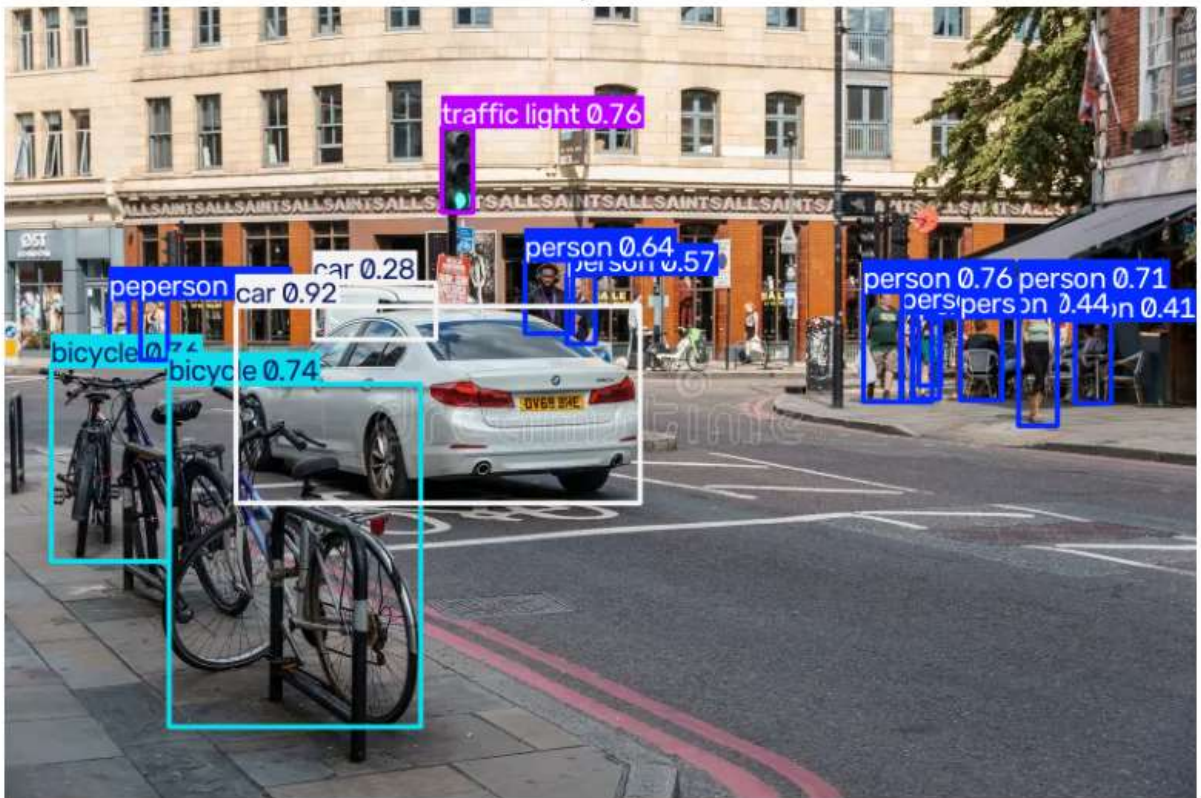
        print(class_name, "Confidence:", round(confidence, 2))

```

Image loaded successfully!

image 1/1 c:\Users\Himanshu\OneDrive\Desktop\CV LAB\london-street-scene-cars-bikes-uk-august-bicycles-people-walking-shops-cafes-sunny-day-445378100.webp: 448x640 11 persons, 2 bicycles, 2 cars, 1 traffic light, 167.7ms
Speed: 13.7ms preprocess, 167.7ms inference, 17.3ms postprocess per image at shape (1, 3, 448, 640)

YOLOv8 Object Detection



Detected Objects:

car Confidence: 0.92

traffic light Confidence: 0.76

person Confidence: 0.76

bicycle Confidence: 0.74

person Confidence: 0.71

person Confidence: 0.64

person Confidence: 0.57

person Confidence: 0.53

person Confidence: 0.44

person Confidence: 0.42

person Confidence: 0.41

bicycle Confidence: 0.36

person Confidence: 0.34

person Confidence: 0.33

person Confidence: 0.32

car Confidence: 0.28

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In [ ]: # Experiment No. 9
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## Experiment Name
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Object Detection and Recognition using Deep Learning Models
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## Aim
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To detect and localize multiple objects in images using a pre-trained deep learning
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## Model Used
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YOLOv8
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## Dataset Used
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Standard Image Dataset
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Objective

To identify objects using bounding boxes, labels, and confidence scores.

QUESTIONS AND ANSWERS

Q1. Differentiate between image classification, object detection, and image segmentation.

Answer: Image classification identifies the main object in an image. Object detection identifies and locates objects using bounding boxes. Image segmentation assigns pixels to different objects or regions.

Q2. Explain the working principle of YOLO.

Answer: YOLO divides an image into regions and predicts object classes, bounding boxes, and confidence scores in a single pass. Therefore, it can perform object detection very quickly.

Q3. Compare YOLO, SSD, and Faster R-CNN.

Answer: YOLO is very fast and suitable for real-time detection. SSD provides a good balance between speed and accuracy. Faster R-CNN generally provides high detection accuracy but is slower.

Q4. What is a bounding box? Why is it important?

Answer: A bounding box is a rectangle drawn around a detected object. It shows the location and size of the object in an image.

Q5. Explain confidence score and IoU.

Answer: Confidence score shows how confident the model is about a detection. IoU measures the overlap between the predicted bounding box and the actual bounding box.

Q6. What are the advantages of pre-trained deep learning models?

Answer: Pre-trained models save training time, require less data, and can provide good detection performance compared with traditional image processing methods.

Q7. Why are MS COCO and Pascal VOC widely used?

Answer: They contain large collections of labeled images with multiple object classes and bounding-box annotations, making them useful for training and evaluating detection models.

Q8. Mention five real-world applications of object detection.

Answer:

1. Autonomous vehicles

2. Security and surveillance
3. Healthcare
4. Retail analytics
5. Smart traffic systems

Q9. What challenges occur in complex real-world environments?

Answer: Common challenges include poor lighting, object occlusion, small objects, different object sizes, complex backgrounds, and crowded scenes.

Q10. How can object detection systems be improved?

Answer: They can be improved using better datasets, data augmentation, advanced YOLO models, transfer learning, hyperparameter tuning, and improved hardware.